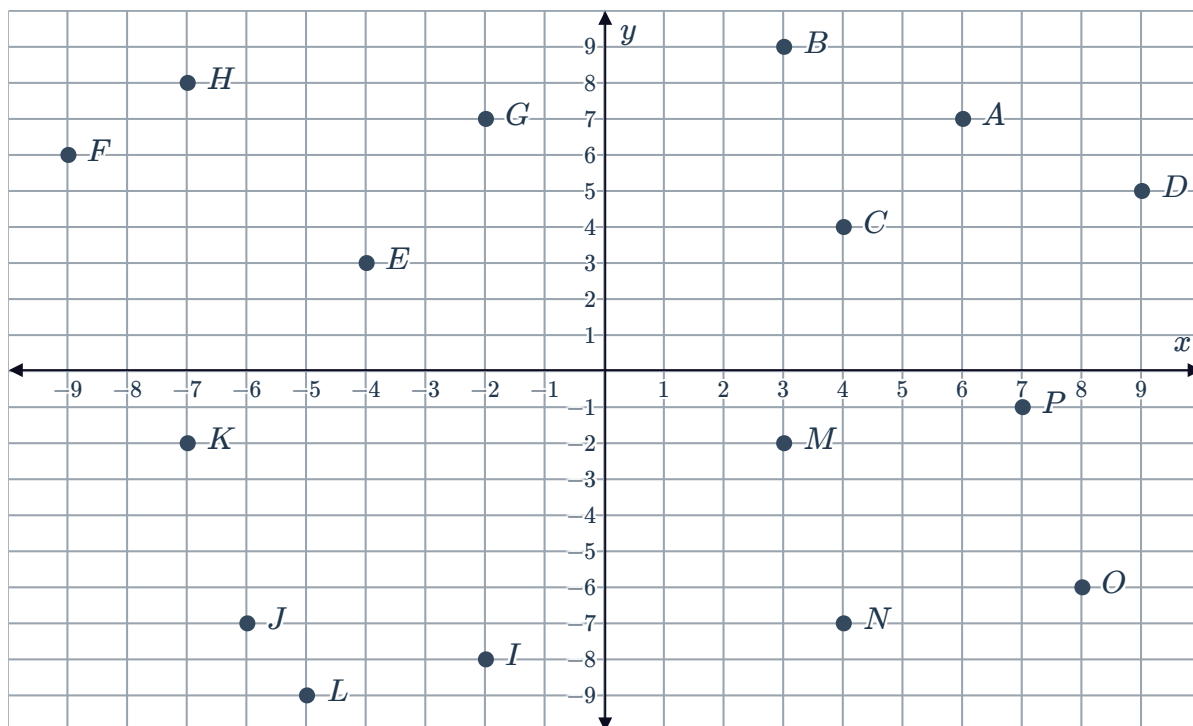


Directed numbers on the number plane



1 Consider the following number plane:



State the coordinates of the following points:

a A

b B

c C

d D

e E

f F

g G

h H

i I

j J

k K

l L

m M

n N

o O

p P

2 Consider the following number plane:

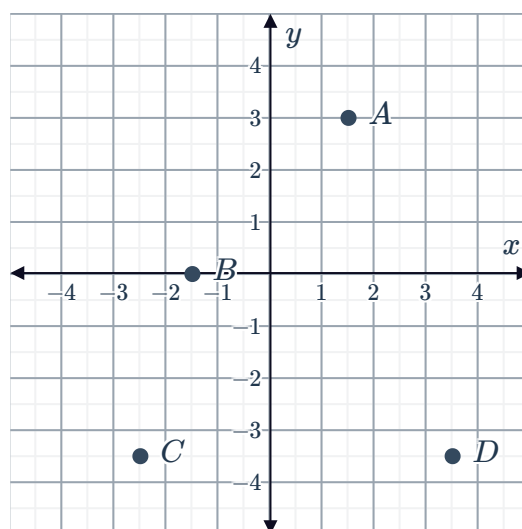
State the coordinates of the following points:

a A

b B

c C

d D



3



a State the x -coordinate of:

i The origin.

ii Any point on the y -axis.

b State the y -coordinate of:

i The origin.

ii Any point on the x -axis.

4 If point P has an x -coordinate of 0, which axis must it lie on?

5 If (a, b) is a point that lies on the y -axis, find the value of a .

6 Consider the given points on the number plane:

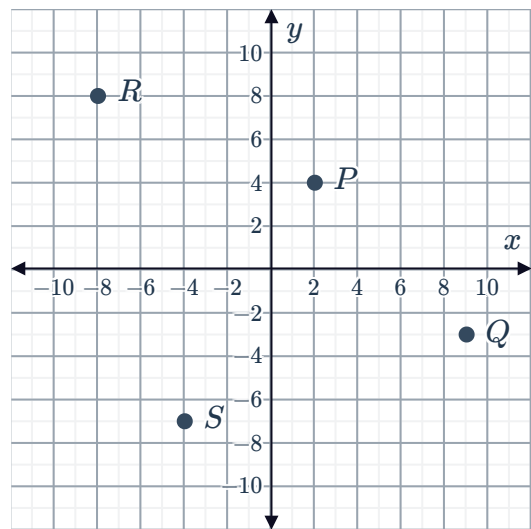
State which point has the following coordinates:

a $(9, -3)$

b $(-4, -7)$

c $(2, 4)$

d $(-8, 8)$

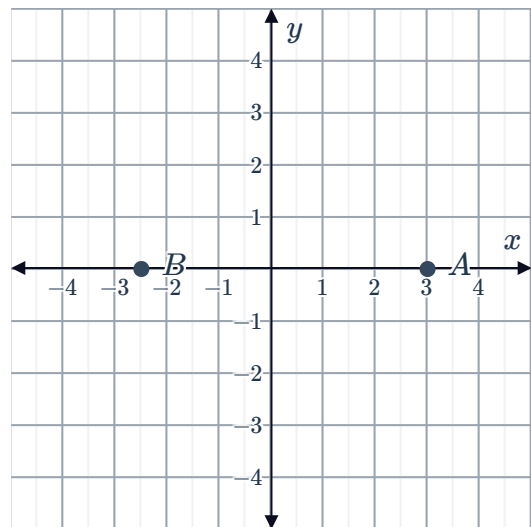


7 Consider the points A and B plotted on the number plane:

a Write down the coordinates of point A .

b Write down the coordinates of point B .

c Which axis do points A and B lie on?



8 In which quadrant do the following points lie?

a $(7, 8)$

b $(-3, 4)$

c $(2, -1)$

d $(-5, -6)$

9 State the quadrant(s) that have:



- a Points with a negative y -coordinate.
- b Points with a negative x -coordinate.
- c Points where the x -coordinate and y -coordinate have the same sign.
- d Points with a negative x -coordinate and a positive y -coordinate.
- e Points with a negative x -coordinate and a negative y -coordinate.

10 Plot the following points on a number plane:

- | | | | |
|---------------|---------------|--------------|---------------|
| a $A(6, 1)$ | b $B(-7, 3)$ | c $C(7, -4)$ | d $D(-7, -3)$ |
| e $E(-4, -4)$ | f $F(-6, 0)$ | g $G(2, 6)$ | h $H(2, -8)$ |
| i $I(-6, 3)$ | j $J(-5, -5)$ | k $K(-5, 9)$ | l $L(5, -9)$ |
| m $M(3, -3)$ | n $N(0, 8)$ | o $O(1, 3)$ | p $P(-4, 5)$ |
| q $Q(-2, -4)$ | r $R(4, -3)$ | s $S(1, 0)$ | t $T(0, 2)$ |

11 For each of the following points:

- | | | | |
|-------------------------------------|--|--------------|-------------|
| i Plot the point on a number plane. | ii State which quadrant the point lies in. | | |
| a $(2, 5)$ | b $(-2, 1)$ | c $(-3, -5)$ | d $(8, -5)$ |
| e $(-8, -1)$ | f $(4, -7)$ | g $(0, 2)$ | h $(5, 0)$ |

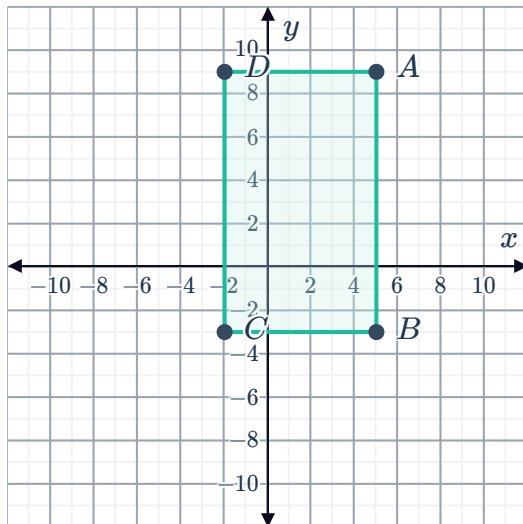
12 For each of the following sets of points:

- | | |
|--|--|
| i Plot the points on a number plane. | ii Identify the shape formed by joining the points in order. |
| a $(4, 1), (7, 1), (4, 4), (7, 4)$ | b $(1, 5), (6, 5), (1, 8), (6, 8)$ |
| c $(-5, -3), (0, -1), (-4, 6), (1, 8)$ | d $(-4, -3), (0, -1), (-2, 1), (2, 3)$ |

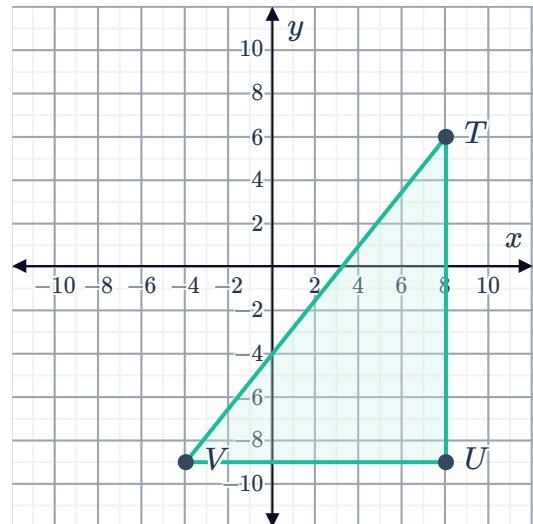
13 For each of the following shapes, write down the coordinates of the vertices:



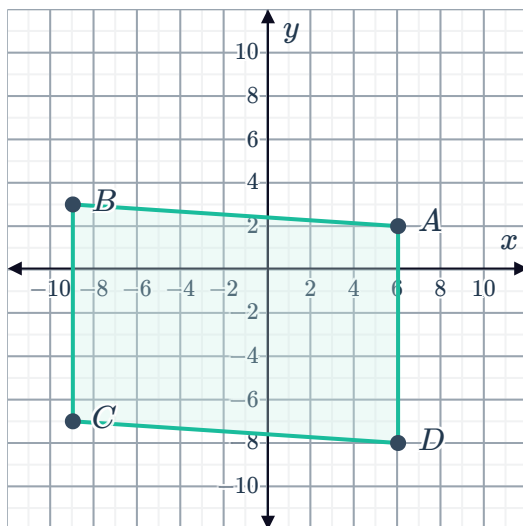
a



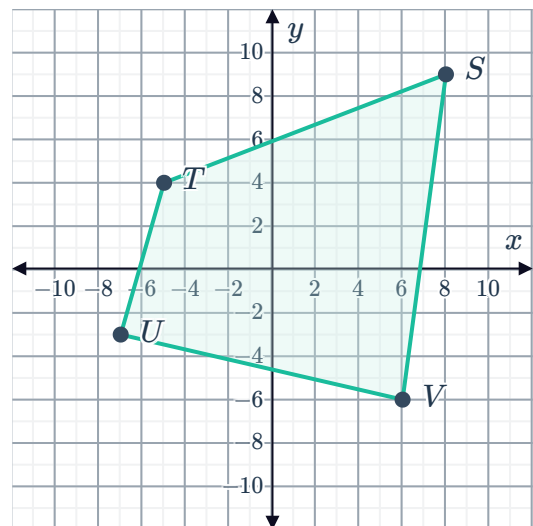
b



c



d



14 Describe how you would move on the number plane from the origin, to plot the following points:

a $(2, -4)$

b $(9, 14)$

15 Find the coordinates of the following points given the description:

a The point 9 units below the origin.

b The point 3.5 units to the left of the origin.

c The point 4 units to the left of $(-3, 6)$.

d The point 7 units to the right of $(-1, -2)$.

e The point 2 units to the right and 2 units below the point $(2, 5)$.

f The point 6 units to the left and 5 units above the point $(4, -4)$.

16 Which point is furthest from the origin?



- A** $(0, -3)$ **B** $(5, 0)$ **C** $(0, 4.5)$ **D** $(-4, 0)$

17 How many units above the origin is the point $(-6, 1)$ located?

18 Starting at the origin, move 4 units left and then 4 units up.

- a** Plot the resulting point on a number plane.
- b** State the coordinates of this point.

Distance between two points

19 Find the distance between the following pairs of points:

- a** $A(-5, 8)$ and $B(-2, 8)$.
- b** $A(7, 3)$ and $B(-1, 3)$.
- c** $A(6, -5)$ and $B(6, -1)$.
- d** $A(-6, 2)$ and $B(-6, -7)$.

20 Consider the points: $A(-4, 8)$, $B(-7, 8)$ and $C(-7, 1)$.

- a** Plot the points on a number plane.
- b** Find the length of AB .
- c** Find the length of BC .

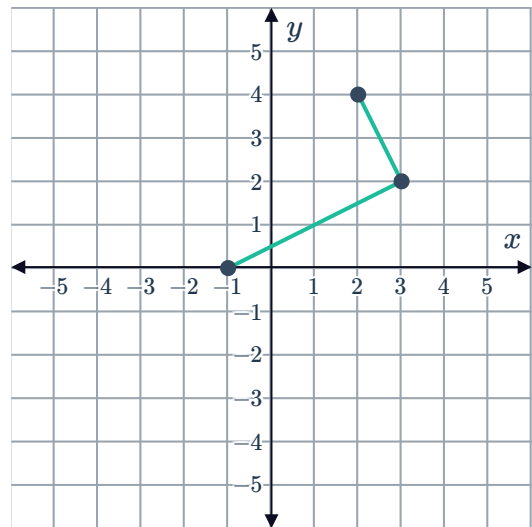
21 A triangle has points $A(1, 2)$, $B(-2, -3)$ and $C(6, -3)$.

- a** Plot the triangle ABC on a number plane.
- b** Find the perpendicular height of the triangle if BC is the base.
- c** Find the length of base BC .
- d** Find the area of triangle ABC .



Geometric applications

- 22** The points given represent three vertices of a parallelogram. Find the coordinates of the fourth vertex if it is known to be in the 2nd quadrant.



- 23** The points given represent three vertices of a rhombus. Find the coordinates of the fourth vertex if the missing point lies in the 3rd quadrant.

